Healthcare Infrastructure Analysis

Abstract:

This project analyzes healthcare infrastructure data using data analytics and visualization techniques to understand the distribution of hospital resources and the level of digital adoption across healthcare facilities. With increasing demand for efficient healthcare systems, data-driven insights can support better planning and allocation of critical resources such as hospital beds, ventilators, and digital health platforms.

The dataset used in this study contains records of healthcare facilities with attributes such as ownership type, number of beds, ventilator availability, 24-hour service status, and ABDM digital enablement. Data preprocessing techniques were applied to clean missing values, standardize categorical labels, and prepare the dataset for analysis.

Exploratory Data Analysis (EDA) was performed using statistical summaries and visualizations to identify patterns among variables. The analysis revealed that hospitals with higher bed capacity generally possessed more ventilators, while ownership type influenced infrastructure distribution. Digital adoption levels also varied across facility categories.

A regression-based predictive model was developed to estimate ventilator availability using hospital characteristics. The findings indicate that infrastructure variables can be useful indicators for resource planning. Overall, this project demonstrates how analytics can support evidence-based healthcare decision-making.

Introduction:

Healthcare infrastructure plays a vital role in ensuring accessible and efficient medical services. Hospitals require adequate physical capacity, emergency support systems, and digital integration to serve growing populations effectively. Resources such as beds, ventilators, and continuous operational support are especially important during public health emergencies.

Recent initiatives promoting digital healthcare systems, such as ABDM, have encouraged hospitals to adopt technology for improved patient records, service

coordination, and healthcare accessibility. Analyzing hospital infrastructure data can therefore provide useful insights into both physical and digital readiness.

Problem Statement:

Despite continued development in the healthcare sector, resource distribution is often uneven across ownership categories and facility types. Some hospitals may have sufficient bed capacity but limited ventilator support, while others may lag in digital adoption.

This project aims to analyze healthcare facility data to understand infrastructure patterns, identify relationships among variables, and develop a predictive model for ventilator availability

Objectives:

The main objectives of this study are as follows:

- To analyze healthcare infrastructure data using statistical methods
- To study the distribution of beds and ventilators across hospitals
- To compare infrastructure based on ownership categories
- To evaluate digital adoption through ABDM enablement
- To build an interactive dashboard using Plotly
- To develop a predictive model for ventilator availability

Data Collection:

The dataset for this project was manually collected from the Ayushman Bharat Health Facilities Registry, which is a trusted national platform containing healthcare facility information.

Initially, automated web scraping was considered for collecting the data. However, due to technical limitations and restrictions on the website structure, scraping was not feasible. Therefore, the required records were manually gathered, verified, and organized into a structured CSV file.

The manual collection process ensured that only relevant and useful attributes were selected for the project. These included:

- Hospital / Facility Name

- Ownership Type
- Number of Beds
- Number of Ventilators
- ABDM Enablement Status
- 24-Hour Operational Availability
- Other facility-level details

Data Cleaning and Preparation:

After collection, the raw dataset required preprocessing before analysis.

The following cleaning steps were performed:

- Missing or incomplete values were identified and handled
- Duplicate records were removed
- Column names were standardized
- Text inconsistencies in categorical values were corrected
- Numerical columns such as beds and ventilators were converted into numeric format
- Data types were validated for accurate analysis

These steps improved the quality and reliability of the dataset.

Exploratory Data Analysis and Visualization:

EDA was carried out using multiple visualizations to understand trends and patterns in the dataset.

Ownership Distribution

The ownership chart showed the share of government and private healthcare facilities in the dataset. This comparison helped in understanding the institutional structure of healthcare services.

Beds vs Ventilators Relationship

The scatter plot between beds and ventilators showed a positive relationship. Hospitals with larger bed capacity generally maintained more ventilators, indicating stronger emergency preparedness.

Learning: Larger institutions are usually better equipped for critical care support.

Top Hospitals by Bed Capacity

A ranked bar chart identified hospitals with the highest number of beds. These facilities likely serve as major healthcare centers in the region.

Learning: A small number of hospitals hold a large share of bed capacity, showing concentration of resources.

Correlation Heatmap

The heatmap was useful in identifying relationships among numerical variables. Bed count and ventilator availability displayed a positive correlation.

Learning: Infrastructure variables are interconnected and can be used for prediction.

Digital Adoption by Ownership

The chart comparing ABDM adoption across ownership types indicated that both government and private hospitals are progressing toward digital integration.

Learning: Digital health adoption is improving across sectors, though some facilities still remain outside the system.

Beds Distribution:

The distribution plot for beds showed that many hospitals have low to moderate capacity, while only a few hospitals possess very high bed counts.

Learning: Healthcare capacity is unevenly distributed.

Key Insights from the Data:

The analysis produced several important findings:

- Hospitals with more beds generally have better ventilator support.
- Resource concentration exists, with a few hospitals dominating bed capacity.
- Ownership type influences infrastructure scale and service availability.
- Both government and private sectors are adopting digital systems.
- Smaller facilities may require greater investment in emergency resources.
- Healthcare infrastructure distribution is not uniform across institutions.

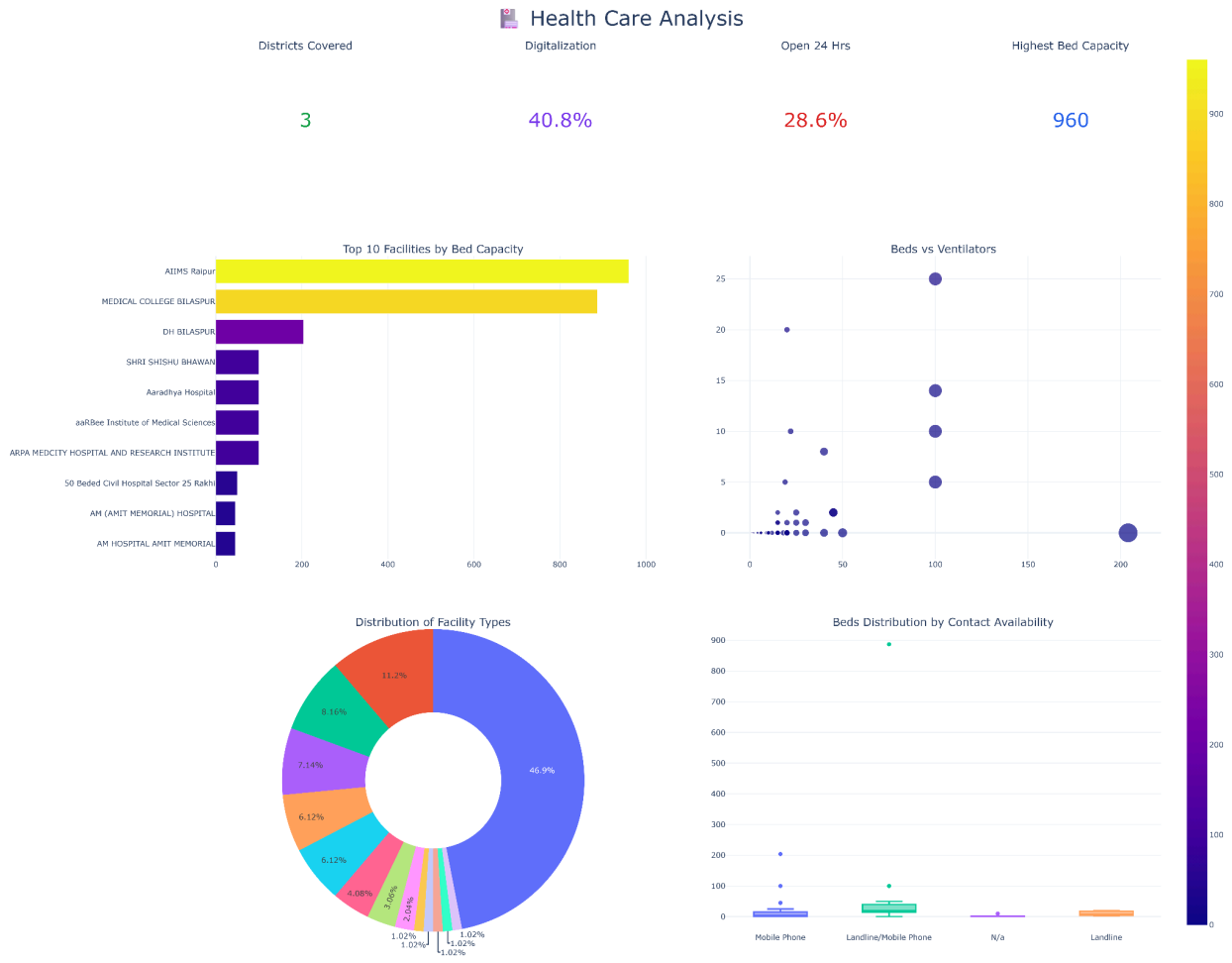
Dashboard Development:

An interactive dashboard was developed using Plotly Dash to make the analysis more accessible and user-friendly.

The dashboard included:

- Districts Covered
- Digitization
- Total Ventilators
- Contact and 24hr Availability
- Beds vs Ventilators Analysis
- Top Hospitals by Capacity
- Digital Adoption Status

This dashboard allows users to interact with the data and instantly observe changes through filters and charts.



Predictive Model:

A regression model was developed to estimate ventilator availability using hospital characteristics.

Features Used

- Number of Beds
- Ownership Type
- 24-Hour Availability

Target Variable

- Number of Ventilators

Linear Regression and Ridge Regression were applied because they are simple, interpretable, and suitable for continuous prediction tasks.

Learning from the Model:

The model indicated that bed capacity is a strong indicator of ventilator availability. Operational and ownership variables also influenced the prediction.

This demonstrates that analytical models can support healthcare planning and infrastructure forecasting.

This project provided practical understanding of the complete analytics workflow, including:

- Real-world data collection challenges
- Importance of cleaning raw datasets
- Choosing suitable visualizations for insights
- Building dashboards for decision support
- Applying machine learning to structured data
- Converting data into meaningful conclusions

The experience also highlighted that collecting original data adds significant value to a project.

Conclusion:

This project successfully analyzed healthcare infrastructure data using data analytics techniques. Starting from manually collected records, the study progressed through cleaning, visualization, dashboarding, and predictive modeling.

The findings showed that hospital capacity, ownership structure, and digital readiness are important dimensions of healthcare planning. Larger hospitals were generally better equipped, while digital adoption was visible across both government and private sectors.

Overall, the project demonstrates how data analytics can be used to improve understanding of healthcare systems and support evidence-based decision-making for future development.

Project_link